

RESEARCH AT THE

Eastern Ohio Resource Development Center



A Branch of the
OHIO AGRICULTURAL RESEARCH AND DEVELOPMENT CENTER
Wooster, Ohio

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FACTS ABOUT THE
EASTERN OHIO RESOURCE DEVELOPMENT CENTER

The Eastern Ohio Resource Development Center, authorized by the 106th Ohio General Assembly in House Bill 949, is operated as one of nine branches of the Ohio Agricultural Research and Development Center at Wooster.

This branch started with the purchase of 728 acres of farm land $1\frac{1}{2}$ miles northeast of Belle Valley in Noble County. The first land purchase was from H. W. and Pearl Morgareidge and was approved by the Center's Board of Control on October 14, 1965. Additional land to make up the 728 acres was purchased from Michael and Thelma Sholtes and Chalmers Wells.

The site was selected because of its central location in the Appalachian Plateau area of Ohio and because of the existence of all major soil types characteristic of Eastern and Southeastern Ohio. A significant factor in the choice of location was proximity of Interstate 77, which will be reached at Belle Valley. Another important factor in the site selection was its location in an area which will not be subject to stripmine operations. Proximity of the Noble County Airport was an added advantage. The above site is known as Unit 1.

During 1966, The Ohio Agricultural Research and Development Center received gifts of 1,000 acres from the Union Carbide Corporation and 325 acres from the Baker and Noon Coal Company. This area is known as Unit 2 of the Eastern Ohio Resource Development Center. It is located approximately 7 miles southeast of Caldwell, with part of it bordering State Route 564. Unit 2 is being used for maintaining part of the 300 beef cattle herd and for stripmine reclamation research and demonstration.

Soils on both units have been mapped in cooperation with U.S. Soil Conservation Service personnel. The soils are quite representative of the soils in southeastern Ohio.

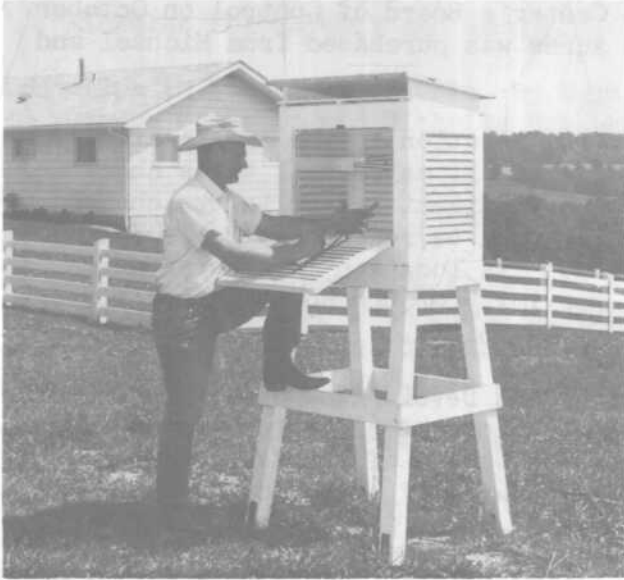
Two residences, one for the manager and the other for the assistant manager, have been completed at Unit 1. A third residence at Unit 2 was remodeled. A 2-mile access road through Unit 1 was constructed. A farm pond comprising 4.1 surface acres, with a storage capacity of 13.8 million gallons or 42.6 acre feet, has been completed. Water from this pond serves four purposes: domestic water supply, livestock, irrigation, and recreation.

The domestic water system includes a concrete block pump house, 8000-gallon raw storage tank, water treatment building with 4000-gallon domestic water supply, sand filter, and chlorinator. A new 30- by 40-foot farm shop with public restrooms was completed in the fall of 1966.

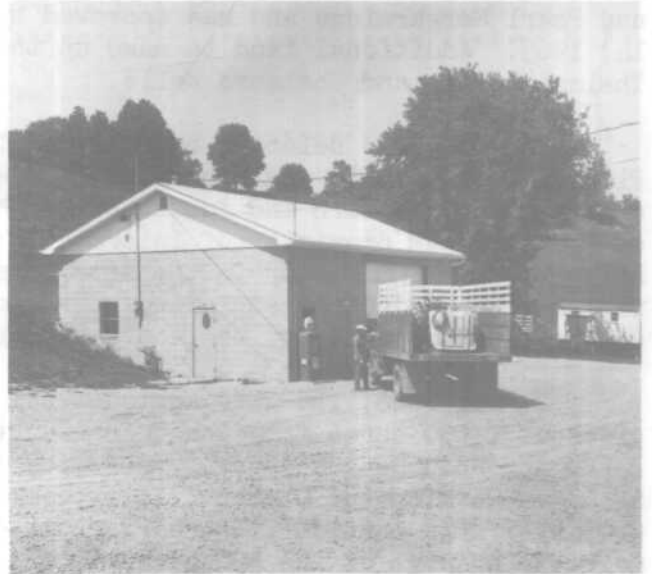
At Unit 2, a 2-mile section of Enoch Township road running through the farm has been widened and stoned through the courtesy of Mr. Howard Devol and the Union Carbide Corporation. An information pavilion, parking area, and observation point were constructed through the courtesy of Mr. Lee Crock and the Crock and Schafer Construction Company. State Route 215 is now being constructed through the northwestern corner of Unit 1. It will be completed soon, making access to the Center more convenient.

Future plans include the construction of an additional residence on the east side of Unit 1. It will serve as headquarters for the beef herdsman. A new 47-by 160-foot poled shelter for sheep will be completed in the fall of 1967. A new 0.4 acre farm pond will be constructed to serve as a water supply for livestock at the north end of Unit 2.

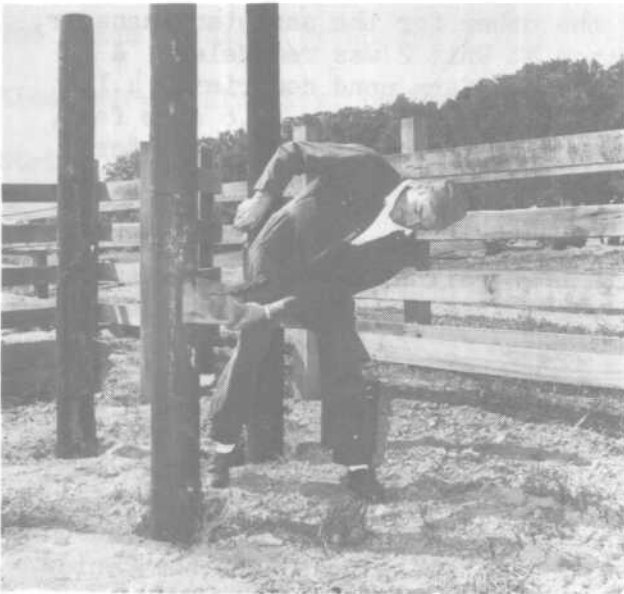
Charles B. Boyles is manager of the Center. Dr. Robert L. Gay, Jr., is in charge of herd health and Dr. Paul Sutton is in charge of stripmine reclamation research.



Manager Charles B. Boyles records daily minimum and maximum air temperatures. Rainfall is also recorded.



New farm shop was completed in the fall of 1966.



James Wells builds cattle corral at Unit 1.



Well-constructed head gate in cattle corral.

SOILS OF THE EASTERN OHIO
RESOURCE DEVELOPMENT CENTER

Nicholas Holowaychuk and George F. Hall
Department of Agronomy

A comprehensive program of soils investigations has been undertaken at the EORDC. One objective of this research is to study the influence of the properties of these soils on the establishment, management, and productivity of forage crops and pasture.

The initial phase of this research was a detailed survey of the soils of both units to show their extent and distribution and to evaluate the degree of erosion in each area. A map of the soils of Unit 1 and a short description of each soil will be prepared during the winter of 1967-68.

Other phases of soils research will involve field and laboratory investigations of the physical, chemical, and mineralogic properties of each major soil occurring on the Center. This basic information will contribute to a better understanding of these soils' properties which influence moisture uptake and availability, ease of working, and liming and fertility needs and responses.

The soils of the Center are similar to those occurring over a considerable part of eastern and southeastern Ohio. Thus, research findings at the EORDC should be of considerable value to a large part of the state.



Soils of Southeastern Ohio will produce good pastures with proper management. On land above, brush was removed and lime and fertilizer added.

LANDSLIPS IN SOUTHEASTERN OHIO

Kaye R. Everett, George F. Hall,
and Nicholas Holowaychuk
Department of Agronomy

Areas of landslide characterized by uneven and irregular microrelief can be seen on many slopes throughout Unit 1. These landslips or slides create problems in tillage, fertilization, and other operations on farms in much of eastern and southeastern Ohio.

Research is directed toward the characterization and identification of landslide areas and the development of methods for stabilization. Measurements are being made of the amount and type of movement on two tracts of land on Unit 1. The physical and chemical characteristics of the soil in the landslide and adjacent areas are being analyzed. After compilation of this data, methods will be proposed for the reduction or elimination of active landslide areas.

On the west portion of Unit 1, an area with characteristic landslide microrelief has been set aside. Sixteen monuments have been established in this area and their exact locations have been recorded. Measurements will be made periodically to determine the amount and type of movement taking place.

Adjacent to this study area, a trench was dug from the bedrock escarpment to a position on the slope below the active landslide zone. The zones of mixing resulting from landslide in the trench were mapped and samples were taken for physical and chemical analyses.

A similar active landslide area has been set aside on the east portion of Unit 1. Monuments will be established, measurements taken, and a trench dug in this area during the fall of 1967 and spring of 1968. In both areas, clay piezometers will be placed at various depths in the study areas to collect data concerning the moisture status in the landslide areas. Correlations will be made between the amount and type of movement which occurs and the moisture content.

BEEF CATTLE BREEDING

Earle W. Klosterman, C. F. Parker, and V. R. Cahill
Department of Animal Science

Income from beef ranks as the second largest source of gross farm income in Ohio. Numbers of dairy cattle have been decreasing and numbers of beef cattle have been steadily increasing. Large land areas of eastern and southern Ohio could be improved to further expand beef production.

A large part of the feed for any animal is required merely to maintain the weight of that animal. The beef industry carries a tremendous overhead in the amount of feed needed to meet the maintenance requirements of breeding herds and stocker cattle. It has been estimated that 75 percent of the feed fed to beef cattle is required for this purpose.

There is a direct relationship between rate of gain and efficiency of gain and also between rate of gain and mature size. Selection for rate of gain will likely increase the mature size of individuals in beef breeding herds. Increased use of large breeds in crossbreeding systems will also bring about a general increase in size of beef breeding cattle. A system of beef production is needed in which a small, highly productive cow will produce a calf of desirable slaughter weight and condition at a young age.

Most efficient gains are generally made by the young calf and milk production of the dam is an important factor in the production of maximum weight at weaning age. More information is needed on the effects of crossbreeding various breeds and sizes of cattle and the importance of milk production of the dam upon the total feed required to produce a unit of edible beef.

There are large differences among the numerous beef and dairy cattle breeds in size, growth rate, milk production, muscling, and ability to marble and fatten. These many breeds have been developed for specific types of production and from a wide genetic background. A number of experiments have been conducted to compare the feed lot performance and carcass traits of some of the beef and dairy breeds. However, controlled experiments in which various breeds of beef and dairy cattle of different sizes have been combined in crossbreeding systems have been very limited.

The objectives of this research are:

1. To study the relationship between age of cow and heterosis of fertility and mothering ability in Hereford, Angus, and Hereford-Angus cross cows.
2. To determine the value of large and small breeds of beef and dairy cattle in a crossbreeding system.
3. To study the effect of size of cattle upon total feed cost per unit of beef produced.
4. To compare various breeds and crosses in their total efficiency of beef production.

Hereford, Angus, and Hereford-Angus cross cows of various ages have been bred to Charolais bulls. The average performance of the Hereford and Angus cows will be

compared to that of the crossbred cows. This will be done on a within-age-of-cow basis in order to obtain information on the first objective of this experiment.

In addition, representative cows of different sizes and breeds mated to bulls of different sizes will be taken to Wooster for a period of 1 year. During this phase of the study, all cows and their calves will be fed individually to measure the total amount of feed required to produce a unit of edible beef.

Steer calves produced at the Eastern Ohio Resource Development Center will be divided into two equal groups. Half of the calves will be sold at weaning time. The other half will be finished on a high energy ration to be hauled into the feed bunks on the stripmine roads of Unit 2. Costs and returns of these two systems of marketing will be compared.



Laverne Wentworth of the farm crew checks beef cattle on pasture.

Hereford, Angus, and Hereford-Angus crossbred cows bred to Charolais bulls are being compared for total efficiency of beef production.



FEEDING VALUE OF DIAMMONIUM PHOSPHATE AND UREA WHEN FED IN A DRY SUPPLEMENT
OR WHEN ADDED TO CHOPPED WHOLE PLANT CORN AT TIME OF ENSILING

Earle W. Klosterman, K. E. McClure, and V. R. Cahill
Department of Animal Science

Research results published previously by the Ohio Agricultural Research and Development Center have shown that the addition of urea to whole plant corn at the time of ensiling is a practical method of increasing the crude protein content of corn silage. It was also found that an addition of 1 percent pulverized limestone or a mixture of equal parts of limestone and urea had a favorable effect upon fermentation in the silo and that silage so treated had a slightly higher feeding value.

When limestone was added at a level of 1 percent of the chopped corn, these additions were approximately 3 percent when calculated on a dry matter basis. Although favorable results were obtained, these additions had a marked effect upon the calcium to phosphorus ratio of the total ration. This raised the question whether or not the feeding results might be improved further by maintaining a more nearly normal ratio of these two major minerals.

Diammonium phosphate (DAP) contains approximately 20% phosphorus and enough nitrogen to supply approximately 106% crude protein equivalent. Thus it is a rich source of phosphorus as well as non-protein nitrogen. The DAP used in this experiment was supplied through the courtesy of the Smith-Douglass Company and contained 21.4% phosphorus, 18.1% nitrogen, and 0.2% fluorine. Fertilizer grades of DAP may contain too much fluorine to be used as a livestock feed.

The silages fed in this experiment were made from well-matured whole plant corn. The amount of materials added at time of ensiling and the dry matter and crude protein analyses of the resultant silages were:

<u>Silage with Additives</u>	<u>Dry Matter Percent</u>	<u>Protein, Percent D.M. Basis</u>
Limestone Silage		
1.0% limestone and 0.1% defluorinated phosphate	35.1	9.2
Urea Silage		
1.0% urea, 0.5% limestone and 0.1% defluorinated phosphate	35.4	14.8
DAP Silage		
1.8% DAP and 0.5% limestone	36.5	12.6

Eighty head of choice quality, Hereford steer calves purchased from a single ranch in northern Texas were used in this experiment. They were shipped to Wooster in early October 1966. However, the supplies of silage were not adequate to finish them to slaughter weight and all steers were fed 8 lb. per head daily of a mixture of 85% ground ear corn and 15% soybean meal with a full feed of hay until January 3, 1967. They were divided at random within weight groups into 8 lots of 10 steers each. All lots were fed in an open sided shed without access to outdoor lots. Each of the following rations was fed to duplicate lots.

TABLE 1.--Feeding Value of Diammonium Phosphate and Urea When Fed in a Dry Supplement or Mixed with Chopped Corn at Time of Ensiling.

	Limestone Treated Silage Plus Dry Corn and Supplement		Supplement Added to Chopped Corn When Ensiled	
	Urea	DAP	Urea	DAP
No. steers	20	20	20	20
Average weight, Jan. 3, lb.	630	628	628	629
Average weight, June 27, lb.	1001	1016	1006	1006
Average daily gain, 175 days	2.12	2.22	2.16	2.10
Average daily ration, lb.				
Silage	29.4	31.4	30.7	30.1
Ground ear corn	6.0	6.0	6.0	6.0
Urea	0.2	-	-	-
DAP	-	.3	-	-
Feed per cwt. gain, lb.				
Silage	1390	1422	1425	1433
Ground ear corn	283	267	274	285
Urea	9	-	-	-
DAP	-	16	-	-
Dressing percentage	60.7	60.6	60.3	60.6
Marbling score*	5.4	5.4	5.0	5.3
Final grade**	18.2	18.4	17.7	18.6
Fat thickness, inches	.40	.44	.42	.48
Kidney, %	2.3	2.2	2.2	2.2
Area rib eye, sq. in.	11.6	11.3	11.1	11.4
Trimmed retail cuts, %	50.9	50.4	50.4	50.3

*Small, 5; modest, 6.

**Low, average and high good 16, 17, 18, respectively.

1. 6.2 lb. per head daily of a mixture of 97.0% ground ear corn and 3.0% urea with the limestone-treated silage full fed.
2. 6.3 lb. per head daily of a mixture of 94.5% ground ear corn and 5.5% DAP with the limestone-treated silage full fed.
3. 6.0 lb. ground ear corn per head daily with the urea-treated silage full fed.
4. 6.0 lb. ground ear corn per head daily with the DAP-treated silage full fed.

In addition, all lots were allowed free access to trace mineralized salt and to a mixture of 4 parts defluorinated rock phosphate and 1 part salt. All steers were implanted with 15 mg. of stilbestrol per head at the start of the experiment.

The experimental rations were fed for 175 days, utilizing all of the silages available. The steers were slaughtered through a local packing plant where individual carcass data were obtained. Average results of this experiment are presented in Table 1.

There was slight variation among lots in average daily gain. However, average gains, feed requirements, and carcass data were very similar whether urea or DAP was fed as the source of nitrogen. There appeared to be little difference whether the urea and DAP were mixed with the silage or with ground ear corn. A similar experiment conducted the previous year and reported in Research Summary 18 indicated that feed requirements per unit of gain may have been slightly lower when DAP was added to the silage. This was not true in the present experiment.

In both of the 1966 and 1967 experiments, very similar results were obtained when urea or diammonium phosphate were used as sources of nitrogen. This was true whether they were mixed with the silage or with dry ground ear corn fed with silage. Thus, the relative cost of nitrogen and phosphorus as supplied by these and other supplements would be a major factor in deciding which to use.

With the age of cattle and length of feeding period used in these experiments, the calcium to phosphorus ratio of the limestone or limestone-urea treated silages does not appear to be critical. Steer performance was not improved by the use of diammonium phosphate, which is a rich source of phosphorus.



A good corral is needed for efficiently handling beef cattle. A cattle oiler in or near corral helps in fly control (lower photo).



YEAR-ROUND PASTURE FOR BEEF COWS

R. W. Van Keuren
Department of Agronomy

Present information indicates that a program of year-round pasture for beef cows can be developed in southeastern Ohio. Such a system would include stockpiling the feed in the field as baled hay and as standing forage. The cattle would remain on pasture the entire year.

Such a program should substantially reduce feed and labor costs and make beef calf production a more profitable enterprise in the hill lands of Ohio. The savings would result from substantially reducing the amount of hay harvested and largely eliminating barn or drylot feeding. There would be little need for manure disposal and bedding.

Southeastern Ohio generally has relatively mild winters, with snow cover of short duration. The average January temperature ranges between 30° and 35° F., depending on location. The hills and wooded ravines can be used to advantage as wind protection for wintering beef cows outdoors.

Aims of the Study

The study with beef cows at Unit 1 of the Eastern Ohio Resource Development Center is aimed at developing a forage system with the cattle remaining on pasture throughout the year. It would include limited mechanical harvesting and no housing except for emergency needs. An objective is to learn if labor and capital outlay can be substantially reduced compared with conventional practices now being followed.

In addition to the cost study, information will be obtained on the effect of such a program on beef cow condition and reproductive efficiency and on calf quality. This phase of the study will determine if supplementation is needed to maintain desirable beef cow condition and to obtain good calf birth and weaning weights.

The forage studies include obtaining information on carrying capacity of such a forage system, forage stand longevity, forage yields, crop sequences best suited to such a program, weed control needed, and lime and fertilizer needs for such a forage system.

Forages Used

It is believed that several species of grass should be used, rather than depending on just one. Currently tall fescue, orchardgrass, and Kentucky bluegrass are the major grasses to be used. Legumes will be included but the greater part of the yield is expected to be from the grasses. Nitrogen fertilization will be used to maintain forage production. Legumes are not dependable in the area and are generally short-lived.

Of the approximately 150 acres in the beef pasture unit, 40 acres are now primarily Kentucky bluegrass. This field has been limed and fertilized to improve the stand and increase production. This field will be utilized primarily for spring and fall grazing.

About 55 acres are now in mixed grasses, primarily orchardgrass and smooth brome grass, with some timothy and reed canarygrass. Most of the field was harvested as hay in 1967. Production from about 10 acres was stored in the barn as emergency feed to be used in case of summer drought, in periods of unusually heavy snow cover in winter, or for other unforeseen needs. Another 35 acres was harvested as round bales and left in the field for winter pasture. This hay, along with the fall regrowth, will be used to winter the herd.

Ten acres of the mixed grass field were left standing for midsummer pasture after the bluegrass pasture was grazed. The cattle were turned on this field in mid-July. The early growth had matured but there was considerable new green growth. The cattle utilized the forage very well, with relatively little waste. After this field was grazed, the cattle were placed on the aftermath of the field harvested earlier for stored feed.

An important aspect of future studies is to learn if maturing pasture is best stockpiled as standing material or as round bales left in the field. Both practices have been followed but good comparative information is lacking.

Winter Pasture

About 50 acres were spring-seeded to Kentucky 31 tall fescue and ladino clover. After this seeding becomes established, it will be used for winter grazing. Hay will be harvested in round bales and left in the field. The baled hay and the accumulated fall regrowth will be used from late November until early spring.

Both Kentucky bluegrass and tall fescue have been used as winter pasture for mature beef cows. Both have been found to be satisfactory for this purpose. Tall fescue is a more productive grass than bluegrass and will carry more animals on a given area than bluegrass.

Year-Round Forage System

The year-round forage system as now planned is based on a sequence of pastures. The cattle will be wintered on the tall fescue. Additional tall fescue will be used for early spring pasture. Kentucky bluegrass will be grazed next. The first cutting of orchardgrass will be stored as hay. The orchardgrass aftermath plus regrowth on the other spring pastures will be used for summer and fall grazing. Following this, the cattle will go back on the tall fescue winter pasture.

The orchardgrass hay will provide the flexibility needed in the system. It can be used to supplement summer pastures in event of unusual drought conditions; during fall, winter, or early spring if extra feed is needed; or in case of unusually deep snow on winter pasture. Surplus hay can be sold.

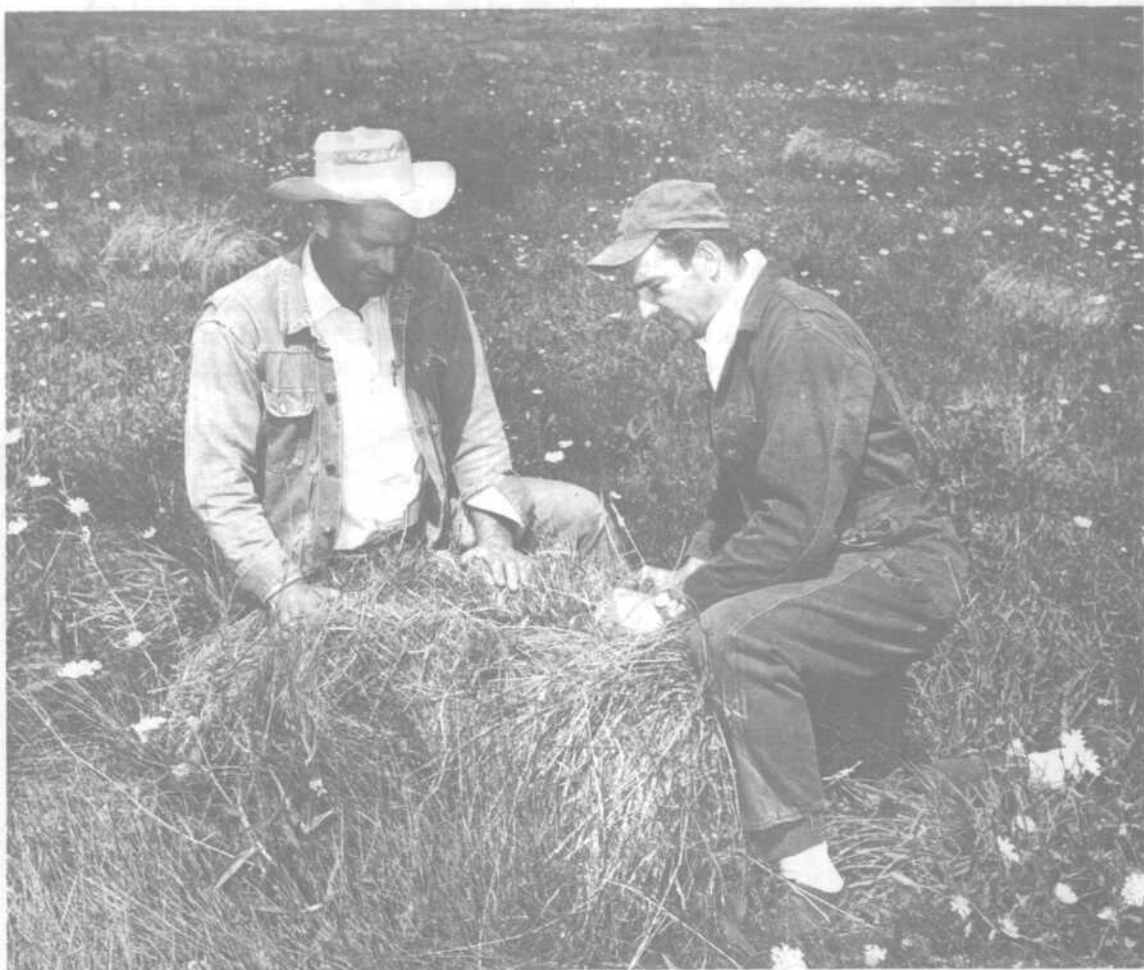
Management Important

The fields should not be overstocked and thus become overgrazed. Continuous overgrazing will reduce forage production, causing further feed shortages. Live-stock numbers must be fitted to the carrying capacity of the forage available on the farm.

The forage management will include applying 300 lb. of ammonium nitrate each spring on the tall fescue and orchardgrass and 200 lb. on the Kentucky bluegrass pastures. Every other fall, 200 lb. of 0-20-20 will be applied to all fields, with

additional amounts if soil tests indicate the need. Liming will be done as needed according to soil test. Such a fertilizer program will give a good return on the money invested. Further study may indicate that greater amounts of fertilizer are economic to use.

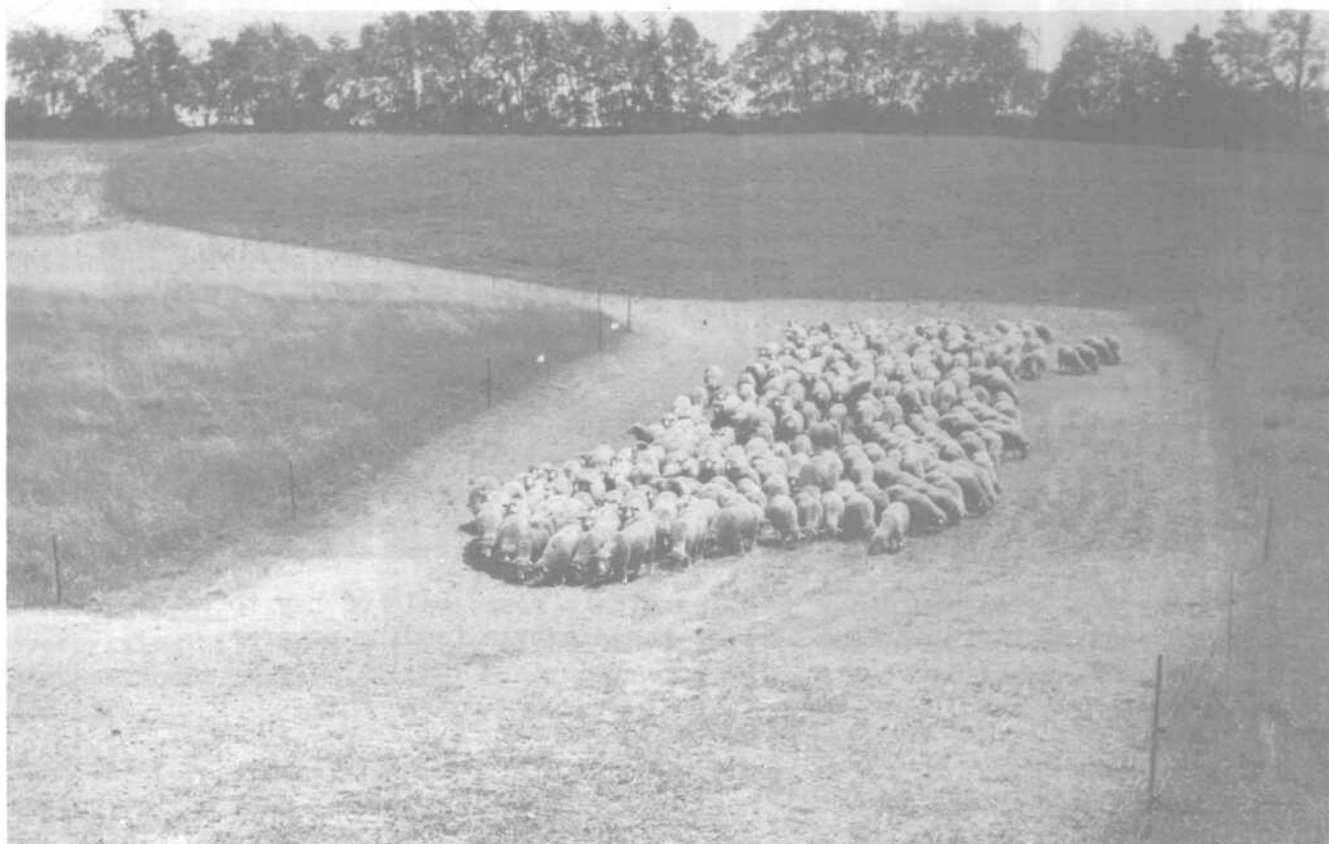
The basic objective of this study is to develop a practical forage system for year-round pasturing of beef cows which can be adapted by any beef producer, whether he has a dozen cows or several hundred. Such a program should result in a substantial reduction in costs and make beef calf production a more profitable enterprise in southeastern Ohio.



Charles B. Boyles and James Wells examine a round bale of hay to be used as supplemental winter feed in year-round cattle feeding program.



Controlled grazing to increase pasture utilization can be achieved by subdividing large pastures with electric fencing and concentrating sheep numbers. This system also permits timely rotation necessary for control of internal parasites. Electricity, an economical method of fencing, has several indirect benefits which influence the net return from sheep.



ELECTRIC FENCING FOR SHEEP

C. F. Parker
Department of Animal Science

A major cost disadvantage to the sheep producer in eastern Ohio has been the expense of materials and labor for fencing sheep. Studies at EORDC during the past year have shown that the minimum cost of building a woven wire fence, including labor, was \$6.30 per rod. Additional concerns to sheepmen of the area have been dog damage, parasite infestation, and other problems related to sheep fencing.

Because of these important economic problems, a study to determine the value of electricity for fencing sheep was initiated at the Ohio Agricultural Research and Development Center during the spring of 1966. Experimental results to date have shown conclusively that electricity can be used efficiently for fencing sheep. The results further indicate that electric fencing has a number of indirect advantages for improving the efficiency of sheep production on Ohio farms.

Proper Conditioning of Sheep

The most important factor in using electric fencing for sheep is the proper initial exposure for successful acclimation of the flock. This can be accomplished as follows:

1. Move the flock into the electrically fenced area slowly to prevent crowding or frightening before the animals have experienced the initial electrical shock.
2. The flock can become acclimated most rapidly to electric fencing if the shepherd can fence one end of a lot where the sheep are confined before going to pasture in the spring. This practice is not essential but is helpful and appears to very effective.
3. Wooden panels or gates should always be used so the sheep can associate areas as the passageways when the flock is moved to another pasture.

Physical Setup

The area where the fence is to be constructed should be mowed or sprayed if the pasture is higher than 10 inches. The most efficient and economical fence to date has been two strands of 14-gauge wire placed on plastic insulators nailed to 2-inch x 2-inch x 4-foot rough wooden stakes approximately 20 feet apart. The heavier gauge wire is important because of the additional strength and ease of handling during the construction or relocation of the fence. A third strand of wire would provide additional insurance against the sheep jumping the fence or putting their heads between the wires. If two strands are used, the bottom wire should be 12 to 15 inches above the ground and the top wire approximately 12 inches higher.

Direct and battery-charged high intensity pulse output fencing units have been used in the studies. Proper grounding of the fencing unit is important, particularly during dry seasons.

Prices on materials such as wire, insulators, and posts are variable and should be studied carefully when deciding the type of materials to use.

Advantages of Electric Fencing

The most apparent advantage of electric fencing for sheep is the low cost of materials and ease of construction. Cost of two 14-gauge wires, insulators, wooden stakes spaced 20 feet apart, the electric unit, and labor is approximately \$1 per rod. This is 15 percent of the cost required to construct a woven wire fence. Two men can construct approximately 1 mile of fence in 6 hours. Less time is required if machinery is used for rolling wire and digging the post holes.

Other advantages of electric fencing may be more important to the sheepmen than the economic saving of construction. These advantages include reduction or complete elimination of dog problems and the use of temporary fencing so corn fields, meadow, and other unfenced areas can be grazed. Large pastures can be subdivided, permitting the concentration of sheep numbers so pastures can be more completely utilized and, with timely rotation, the harmful effects of internal parasites prevented. The condition or state of fatness can be controlled more easily when the owner has more pastures and better control over the grazing habits of the sheep.

It is recommended that sheepmen strongly consider the use of electric fencing for increasing the net return from sheep on Ohio farms.



Rambouillet, Targhee, and Western Blackface crossbred ewes comprise the initial sheep flock at Unit 1.

SHEEP RESEARCH FOR EASTERN OHIO

C. F. Parker
Department of Animal Science

An intensive sheep research program has been planned for Unit 1 of the EORDC. A study of the existing sheep industry and the available farm resources indicates that research investigations are essential to more objectively identify those factors and practices necessary for establishing a profitable sheep enterprise in eastern Ohio. The economic potential of sheep for the production of quality lamb meat and wool suggests that current returns per acre in the area can be more than doubled.

The general areas identified for research study include proper choice of breed type, mating system, pasture program, and management system.

Research Objectives

The following objectives relate the specific kinds of sheep research which have been planned.

1. To determine the type of ewe which will be most productive under the varying environmental conditions existing in the area.
2. To compare pure breeding and crossbreeding as mating systems for the commercial production of lamb and wool.
3. To develop and compare management systems for intensive and extensive sheep production. This approach will evaluate the following components of sheep production management:
 - a. Kinds of pastures for increasing the carrying capacity. Pasture species and methods of pasture management will be evaluated.
 - b. Forage program for extending the normal grazing season.
 - c. Methods of winter feeding. Winter grazing and restricted self-feeding programs will be studied.
 - d. Conventional vs. limited housing.
 - e. Types of fencing, i.e. temporary vs. permanent.
4. To study values of early, late, and multiple lambing systems. The following management treatments will be evaluated.
 - a. Creep feeding.
 - b. Early weaning.
 - c. Finishing lambs on forage.
5. To determine the value of companion grazing (sheep and cattle) for maximum returns from the utilization of pasture and forage crops.

6. To determine the costs of the inputs for each management alternative listed in the preceding objectives, to determine the most profitable and efficient practices, and to reveal the optimum economic production management techniques.

A population of 350 breeding ewes of the Dorset, Rambouillet, and Targhee breeds and their crosses will be established to begin the basic research plan. Rambouillet, Targhee, and Western Blackface crossbred ewes are available for research at present. A structure for housing sheep with working pens will be constructed within the next year. The research at this Center has been designed and will be interpreted to answer questions pertinent to the sheep industry of eastern Ohio.

APPLE CULTIVAR EVALUATION



Excellent orchard sites in Southeastern Ohio have good air drainage and soil conditions for fruit production.

APPLE CULTIVAR AND ROOTSTOCK EVALUATION TRIALS
FOR SOUTHEASTERN OHIO

Freeman S. Howlett
Department of Horticulture

Selection of the most profitable cultivar (variety) is a major requirement for success of the apple industry. Tree size is also particularly important today since large trees requiring tall ladders are no longer desired. Furthermore, high yields per acre are necessary if the enterprise is to be profitable.

The present apple orchard was established at Unit 2 of EORDC in the spring of 1965 to ascertain the relation of cultivar and rootstock to the growth and yield of trees under the prevailing environmental conditions. Fifteen cultivars have been established on the semi-dwarfing rootstock Malling-Merton 106 and the very dwarfing Malling 26.

The plots of trees on Malling-Merton 106 and Malling 26 are planted alternately. Each plot is comprised of the 15 cultivars planted at random within the plot in accordance with accepted statistical procedures.

The main characteristics of the two stocks are as follows:

Malling-Merton 106: Trees on this stock usually bear their first flowers 4 years after planting. The size of the trees at maturity depends on several factors such as amount of fertilizer applied, amount of pruning, soil management system utilized, available water supply and annual yield. If the trees are stimulated to make maximum growth, at 25 years of age they may reach 2/3 to 3/4 the size of trees on standard non-dwarfing rootstocks. Determining how large they will grow on the present location is one objective of the experiment. This information is valuable in indicating optimum planting distance and consequently total yield.

Malling 26: This is a new rootstock released in England in 1958. The trees in this orchard are the first ones propagated on this rootstock in any experimental planting in Ohio. The precise size of trees at 10 to 20 years of age obviously is unknown but it is believed that they will become less than half the size of those on Malling-Merton 106. Possibly this rootstock may be the most suitable for commercial plantings in Ohio in the future. This experiment in Southeastern Ohio will provide very important information on this point.

The cultivars established on these two rootstocks are:

Melrose, Starking, Chelan Red, Jonathan, Red Prince, Idared, Ruby, Gallia Beauty, Holiday, Close, Melba, Golden Delicious, Sundale Sturdy Spur, Delicious, Rogers McIntosh, and Franklin. Several other cultivars have been planted at the summit of this orchard. These are Hi-Early, Red Delicious, Skyline Supreme, Wallspur, Red King, Top Red, Royal Red, Redspur, and Starkrimson.

The trees are planted on the contour, with 18 feet between rows. The distance between trees on Malling 26 is 12 feet and the distance between trees on Malling-Merton 106 is 18 feet.

A few flowers bloomed on some trees in 1967. Presumably considerable flower production will occur in 1968 on these trees, which are situated on one of the best sites for an orchard in Southeastern Ohio. Excellent growth has occurred and the entire orchard soon should be bearing very satisfactorily.

GRAPE RESEARCH IN SOUTHEASTERN OHIO

Garth A. Cahoon
Department of Horticulture

The initial grape planting was established at Unit 1 of the EORDC in the spring of 1966. It consists of a small group of eight selected cultivars plus a block of vines of the cultivar Buffalo. This latter cultivar will receive various nutritional treatments of nitrogen and potassium, commencing in the spring of 1968.

In the spring of 1967, a 1-acre grape planting was established to investigate the interrelationships of irrigation, trellising, and fertility. Treatments will be initiated in the spring of 1968. Soil conditions at this site should be especially adapted to irrigation studies because of the hard-pan condition underlying the area which will restrict root growth.

Three cultivars are being used in the experiment: Seibel 5279, a hybrid which is especially suitable as a commercial wine grape; the standard eastern grape, Concord; and a Missouri hybrid, Blue Eye. The latter has not been fully evaluated as yet but shows promise because of the very large and productive clusters which it produces.

Trellising and training of grapes is currently receiving considerable attention by research laboratories throughout the eastern United States. One of the principal efforts of the grape research program at this Center will be directed toward producing high vigor vines by irrigation and fertility and at the same time achieving maximum productivity and quality by adequate trellising and training techniques.



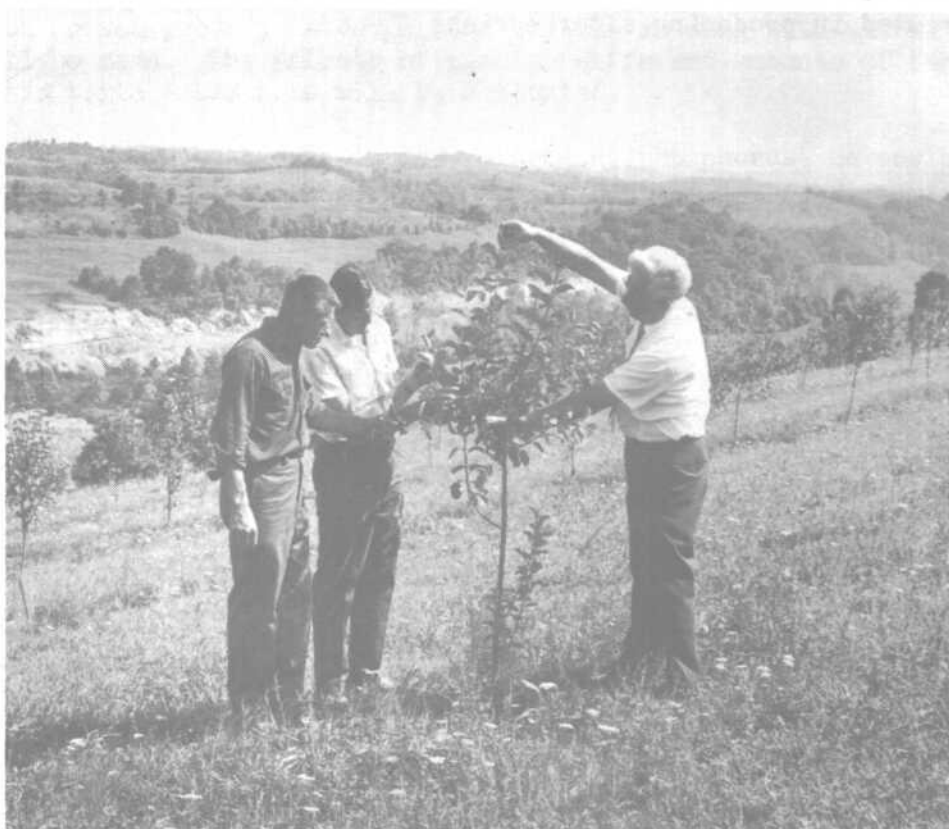
A total of $1\frac{1}{2}$ acres of grapes have been planted at Unit 1. Varieties are being evaluated as part of research program to assist in re-establishment of the grape industry in Southern Ohio.

THE PEACH ORCHARD

Robert G. Hill, Jr.
Department of Horticulture

The peach cultivar orchard was established at Unit 2 of EORDC in April 1967. The main test consists of 16 cultivars in replicated plots. This group includes 10 clingstone cultivars which are particularly well suited for the manufacture of baby foods. The rest of these plots are planted to freestone types known to be well suited to the fresh fruit trade. In addition, the orchard contains 11 other cultivars which were planted for preliminary observational purposes. This group includes four nectarines or fuzzless peaches. All trees are set in three-tree plots. Each tree is labeled as to cultivar and date of planting.

The orchard is being maintained according to recommended horticultural practices and recommended programs for disease and insect control. Data will be obtained on the growth and yield responses of these cultivars to local soil and weather conditions. The results will serve as a basis of recommendations to growers interested in developing a peach industry in the area.



New growth in apple orchard at Unit 2 is examined by Robert Schehl of the farm crew; Dr. Paul Sutton, agronomist in charge of stripmine reclamation research; and T. F. Wonderling, coordinator of research operations. Note the stripmine area in the background.

ELDERBERRY EVALUATION

Robert G. Hill, Jr.
Department of Horticulture

Elderberries have been popular in Ohio for many years. They are used for pies, jellies, jams, and wine-making in Ohio homes and by commercial processors. The demand for these berries has traditionally been satisfied by the "wild" crop. The wild crop, however, apparently can no longer be relied upon to satisfy the demand. As a result, renewed interest is being generated in producing elderberries as a cultivated crop.

To ascertain the feasibility of producing elderberries as a commercial crop in southeastern Ohio and to determine their cultural requirements, a small planting was established at Unit 2 of the EORDC in April 1967. The planting consists of 10 replicated plots of three plants each of six different cultivars. These were: Adams #1, Adams #2, Johns, New York 21, Scotia, and York. All have been reported to have desirable growth, yield, and berry characteristics.

Data will be collected on various growth and yield characteristics under local environmental conditions and observations on cultural requirements will be made. The results obtained will serve as the basis for recommendations to growers interested in producing elderberries.



Robert Schehl examines first fruit on elderberry planting at Unit 2.

STRIPMINE RECLAMATION RESEARCH PLANS
FOR UNIT 2

Paul Sutton
Department of Agronomy

Stripmine reclamation research at the EORDC will be concerned with eliminating toxic areas in the spoil banks, seeding and planting methods, establishing fertility requirements for existing forages, and evaluating different kinds of vegetation on soil formation.

In the elimination of toxic areas, studies will be conducted to determine the effects of different methods of grading and surface preparations on the total soluble salts and the acidity of the areas. An evaluation will be made of effects of surface preparations on water infiltration. These studies will be conducted with a sprinkling infiltrometer. Other studies will be made to determine the amount of top soil which will have to be placed on the spoil bank to establish vegetation, the effects of different amounts of limestone, additional grading to bury toxic materials, and mixing of the spoil material by deep plowing.

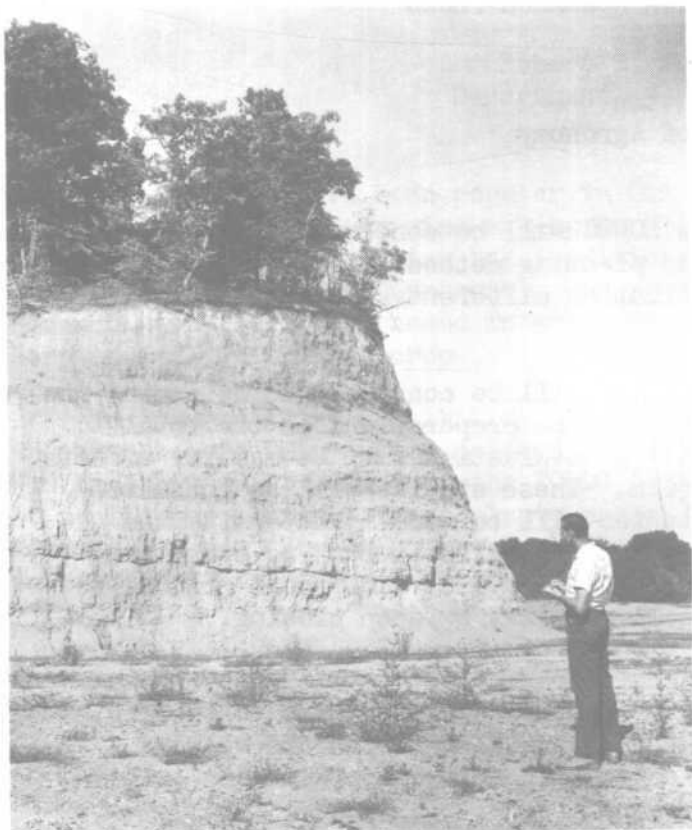
To evaluate seeding and planting methods, forages will be seeded by broadcasting with no surface preparation prior to seeding, disking prior to seeding, and seeding on frozen ground. A comparison of band seeding with and without limestone will be made. The effects of placing different amounts of top soil around trees in toxic areas also will be evaluated.

After forages have been established, nitrogen, phosphorus, potassium, and lime requirements for optimum production will be evaluated.

The effects on soil development of forages and forage and tree combination plantings will be evaluated.



This water impoundment is not contaminated by acid water from stripmine area and thus is good livestock water source.



Dr. Paul Sutton examines soil layers in highwall (left). Stripmined area below is graded to bury toxic materials.



The State Is the Campus for Agricultural Research and Development



Ohio's major soil types and climatic conditions are represented at the Research Center's 11 locations. Thus, Center scientists can make field tests under conditions similar to those encountered by Ohio farmers.

Research is conducted by 13 departments on more than 6000 acres at Center headquarters in Wooster, nine branches, and The Ohio State University.

Center Headquarters, Wooster, Wayne County: 1918 acres

Eastern Ohio Resource Development Center, Caldwell, Noble County: 2053 acres

Mahoning County Experiment Farm, Canfield: 275 acres

Muck Crops Branch, Willard, Huron County: 15 acres

North Central Branch, Vickery, Erie County: 335 acres

Northwestern Branch, Hoytville, Wood County: 247 acres

Southeastern Branch, Carpenter, Meigs County: 330 acres

Southern Branch, Ripley, Brown County: 275 acres

Vegetable Crops Branch, Marietta, Washington County: 20 acres

Western Branch, South Charleston, Clark County: 428 acres